

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

Angela Zavaleta Bernuy



Last Lecture

- Algorithm is a finite sequence of precise instructions for solving a real-life problem.
- Pseudo-code is a translation of an algorithm into a specific programming language.
- Pseudo-code for finding a maximum in a list
- Pseudo-code for linear search
- Pseudo-code for binary search
- Pseudo-code for bubble sort and insertion sort

Sec 3.2: Growth of functions

- How do we compare two different algorithms?
- How do we quantify the efficiency of an algorithm?
- View an algorithm as a function of its input (or the size of the input). We wish to develop a framework for quantifying the growth of a function, i.e., how fast $f(n)$ increases when we increase n .
- This framework quantify growth of f by comparing it with the growth of some easy functions using the following notation:

$$O, o, \Omega, \omega, \Theta$$

Big-O notation

Definition

Let f and g be two functions $\mathbb{R} \rightarrow \mathbb{R}$. We say that $f(x)$ is $O(g(x))$ if there exists constants C and k such that

$$|f(x)| \leq C \cdot |g(x)|$$

for all $x \geq k$.

- Intuitively, $f(x)$ is $O(g(x))$ means f grows slower than a constant multiple of g asymptotically
- Some textbooks write $f(x) = O(g(x))$ instead of " $f(x)$ is $O(g(x))$ "

Example

$$|f(x)| \leq C \cdot |g(x)|, \quad x \geq k$$

- Let $f(x) = x^2 + 2x + 1$ and $g(x) = x^3$. Show that $f(x)$ is $O(g(x))$. , $x \geq 1$

$$\underline{x^2 + 2x + 1} \leq \underline{x^2} + \underline{2x^2} + \underline{x^2} = 4x^2$$

$$x^2 \leq x^3 \quad |f(x)| \leq 4x^2 \leq \underline{\underline{4x^3}}, \quad C=4$$

- Let $f(x) = x^2 + 2x + 1$ and $g(x) = x^2$. Show that $f(x)$ is $O(g(x))$.

$$\Rightarrow x^2 + 2x + 1 \leq x^2 + 2x^2 + x^2 = 4x^2$$

$$C=4$$

Example

We just proved that

- $x^2 + 2x + 1$ is $O(x^3)$
- $x^2 + 2x + 1$ is $O(x^2)$ Which one is a stronger result?

looser bound

Big-O - upper bound.

- Can x^3 be $O(x^2)$? No.

if true. x^3 would eventually
be bounded by x^2 .
(not possible)

A General Theorem

Theorem

Let f be a polynomial function of degree n , that is,

$$f(x) = a_n \underline{x^n} + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0,$$

for some constants $a_n, a_{n-1}, \dots, a_1, a_0$. Then, we have $f(x) = O(x^n)$.

last example, $n=2$, $f(x) = x^2 + 2x + 1$, $\overrightarrow{O(x^2)}$

if $x \geq 1$

for every. $0 \leq m \leq n$, $x^m \leq x^n$

Example

$$x \geq 1$$

- $x^5 + 2x + 1$ is $O(x^?) \rightarrow O(x^5)$
- $x^5 + 2 \log x + 1$ is $O(x^?) \rightarrow O(x^5)$
- $\frac{x^5 + 2 \log x + 1}{x^3 + 2x + 1}$ is $O(x^?) \rightarrow O(x^2)$

$$\text{ratio} \sim x^{5-3} = x^2$$

Special functions

- $n!$ is $O(n^n)$

$$n! = n \times (n-1) \times \dots \times 2 \times 1$$

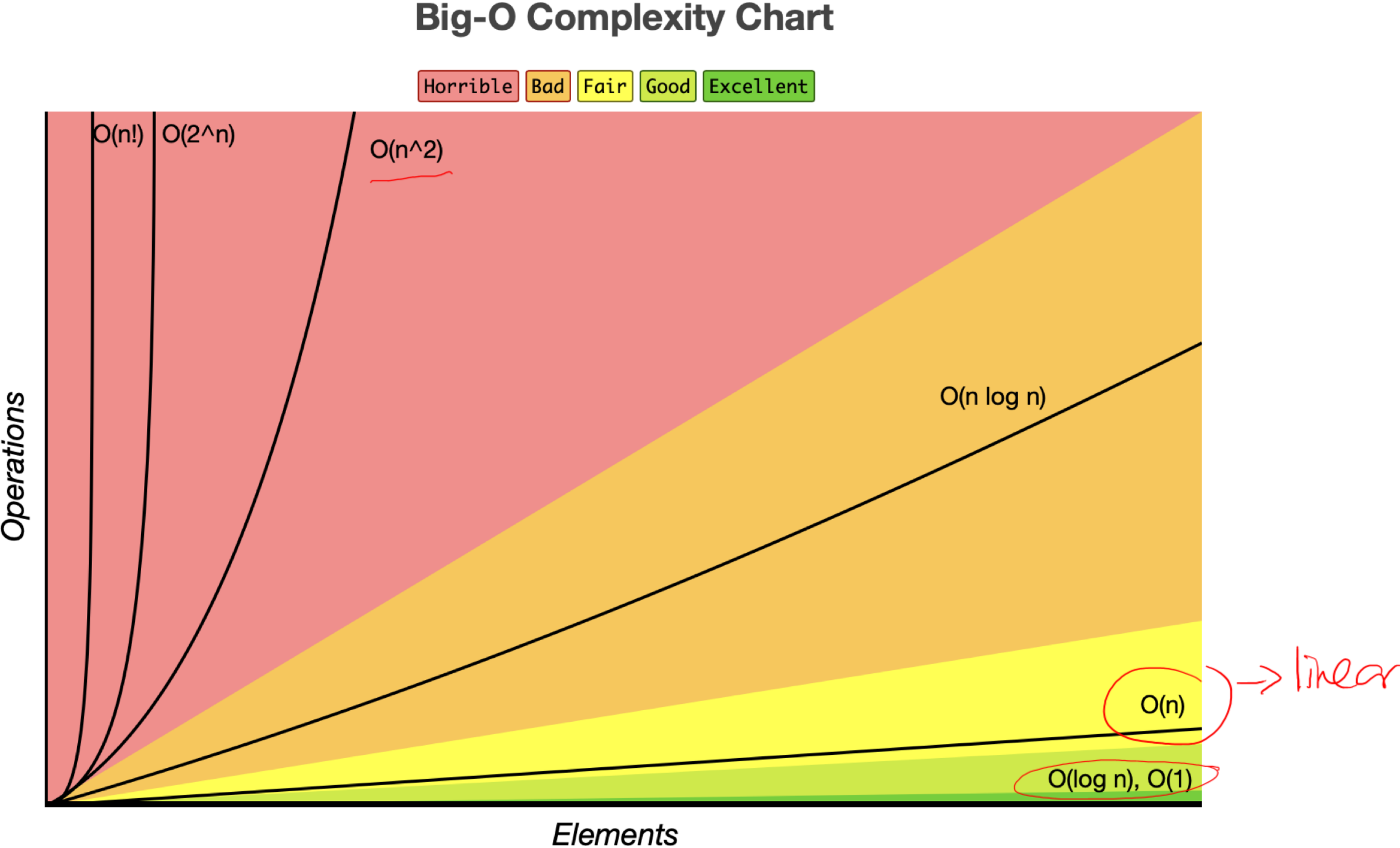
$$n! \leq n \times n \times \dots \times n \times n = n^n$$

- For any constants $b, c \geq 1$, n^b is $O(c^n)$
- For any constant c , c^n is $O(n!)$
- The big-O order of some functions

$$|f(x)| \leq C \cdot \lg(x)$$

$$\underline{1}, \log n, n, n \log n, n^2, 2^n, n!$$

Big-O Functions



Ω -notation

Definition

Let f and g be two functions $\mathbb{R} \rightarrow \mathbb{R}$. We say that $f(x)$ is $\Omega(g(x))$ if there exist constants C and k such that

$$|f(x)| \geq \underbrace{C \cdot |g(x)|},$$

for all $x \geq k$.

- Intuitively, $f(x)$ is $\Omega(g(x))$ means f grows faster than a constant multiple of g **asymptotically**
- $f(x)$ is $\Omega(g(x))$ if and only if $g(x)$ is $O(f(x))$.

Example

- Show that $x^5 + 2x + 1$ is $\Omega(x^5)$.

$$\underline{x^5 + 2x + 1 \geq x^5}$$

$$|f(x)| \geq C \cdot \lg(x) \quad ; \quad \underline{\text{for all } x \geq 1}$$
$$C = 1$$

- Show that x^n is $\Omega(x^m)$ for $m \leq n$.

$$\underline{x^n = x^{n-m} \cdot x^m} \geq \overbrace{x^{n-m} \geq 1} \cdot x^m = x^m, \quad C = 1$$

- Show that x is $\Omega(\log x)$

$$\underline{x \geq \log x}$$

$$C = 1$$

$$\text{for all } x \geq K = 2$$

Θ -notation

Definition

Let f and g be two functions $\mathbb{R} \rightarrow \mathbb{R}$. We say that $f(x)$ is $\Theta(g(x))$ if $f(x)$ is both $O(g(x))$ and $\Omega(g(x))$. Equivalently, $f(x)$ is $\Theta(g(x))$ if there exist constants C_1, C_2, k such that

$$\underbrace{C_1}_{\text{red circle}} \underbrace{|g(x)|}_{\text{red underline}} \leq |f(x)| \leq \underbrace{C_2}_{\text{red circle}} \underbrace{|g(x)|}_{\text{red underline}}$$

for all $x \geq k$.

$$x^2 \leq |x^2 + 2x + 1| \leq 4|x^2|$$

- When $f(x)$ is $\Theta(g(x))$, we say that $f(x)$ is of order $g(x)$, or that $f(x)$ and $g(x)$ have the same order.

- Example: $x^2 + 2x + 1$ is of order x^2 . $\rightarrow \Theta(x^2), O(x^2), \Omega(x^2)$

$$x^2 + 2x + 1 \leq x^2 + 2x^2 + x^2 = 4x^2$$

$$x^2 + 2x + 1 \geq x^2$$